

A Discrete Hawking Quantum from σ -Pair Excitations on the Binary Icosahedral Group

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Abstract

We define the σ -Galois pair-excitation observable $E(v) := |v - \sigma(v)|_{\text{Eucl}}^2$ on the binary icosahedral group $2I = V_{600}$, where $\sigma : \sqrt{5} \mapsto -\sqrt{5}$ acts componentwise on icosian quaternions. We prove that $E(v) = 0$ for the 24 σ -fixed vertices (the 24-cell sublattice V_{24}) and $E(v) = 5/2$ *exactly* for each of the remaining 96 σ -mobile vertices. The numerical value $5/2$ follows from a direct coordinate calculation; constancy of E on the σ -mobile sector additionally follows from a symmetry-grade argument: the bilateral action of V_{24} , $v \mapsto gvh$ for $g, h \in V_{24}$, is transitive on σ -mobile vertices and commutes with σ . Combined with the Paper 1 [Smart, 2026] per-coset incidence $|gH \cap V_{24}|/|gH \setminus V_{24}| = 4/16 = 1/4$ for $H = \text{Dic}_5$, this yields an exact discrete first-law identity per coset (left or right, bulk or non-bulk): $E_C = T_{\text{casc}} \cdot A_C$, $S_C = A_C/4$, $T_{\text{casc}} \cdot S_C = E_C/4$, with $T_{\text{casc}} = E_q = 5/2$ in Euclidean cascade units and $A_C = 16$, $S_C = 4$, $E_C = 40$. Interpreting σ -mobile vertices as Hawking radiation channels and σ -fixed vertices as horizon anchor states gives a discrete monochromatic analogue of Hawking radiation. The semiclassical recovery of a Planck continuum from the discrete gap and the calibration of E_q to $\hbar c^3/(8\pi G M k_B)$ are stated as Tier-2 work.

1 Introduction

The Hawking temperature $T_H = \hbar c^3/(8\pi G M k_B)$ [Hawking, 1975], in conjunction with the Bekenstein–Hawking entropy $S_{\text{BH}} = A/4$ in natural units [Bekenstein, 1973], defines a thermodynamic structure for black-hole horizons. In the semiclassical formulation the radiation spectrum is approximately Planckian: the horizon emits a continuum of modes with thermal occupation at temperature T_H . The continuum nature is fundamental in QFT in curved space-time.

This paper reports a structurally distinct picture: in the finite arithmetic of the binary icosahedral group $2I$ — equivalently the 120 vertices of the 600-cell V_{600} — the natural σ -pair excitation observable yields a single quantum gap $E_q = 5/2$ shared by all σ -mobile vertices, not a continuum. Constancy of the spectrum on the σ -mobile sector is explained by a transitive symmetry — the bilateral action of the σ -fixed sublattice V_{24} on V_{600} — and the numerical value is computed in coordinates. Combined with the Paper 1 [Smart, 2026] coset-incidence identity $S/A = 1/4$, this produces an exact discrete first-law analogue $T_{\text{casc}} \cdot S = E/4$ with $T_{\text{casc}} = E_q$.

The paper does not derive physical Hawking radiation. It claims a finite-arithmetic identity in V_{600} that has the algebraic shape of a discrete Hawking quantum, with the semiclassical recovery of the Planck continuum and the calibration to physical units stated as future work. Section 2 fixes notation and imports the $V_{600}/\text{Dic}_5/V_{24}$ structure from Paper 1. Section 3 defines the σ -pair excitation observable. Section 4 states and proves the monochromatic-spectrum theorem. Section 5 derives the per-coset first-law identity. Section 6 sets the scope.

2 Setup

We import the following from Paper 1 [Smart, 2026]; the present paper does not re-prove any of these statements.

- $K := \mathbb{Q}(\sqrt{5})$ with Galois twist $\sigma : a + b\sqrt{5} \mapsto a - b\sqrt{5}$, extended componentwise to the icosian quaternion algebra $\mathbb{H}_K$.
- $V_{600} = 2I \subset \text{Sp}(1, K)$: the 120 unit icosians forming the binary icosahedral group, partitioned into 8 axis, 16 half-type, and 96 even-permutation vertices.
- $V_{24} := \{v \in V_{600} : \sigma(v) = v\}$: the σ -fixed sublattice, equal to the 8 axis vertices plus the 16 half-type vertices and forming the regular 24-cell. This is also $V_{600} \cap \sigma(V_{600})$.
- σ -mobile vertices: $V_{600} \setminus V_{24}$, the 96 even-permutation vertices.
- $H := \text{Dic}_5 \subset 2I$: the bulk subgroup of order 20, the union of the $K = 72$ and $K = 0$ cycles of T_7 . H is self-normalizing in $2I$.
- Per-coset incidence (Paper 1, Theorem on coset incidence): for every left coset gH and right coset Hg , $|gH \cap V_{24}| = |Hg \cap V_{24}| = 4$ and $|gH \setminus V_{24}| = |Hg \setminus V_{24}| = 16$.

3 The σ -pair excitation

Definition 3.1 (σ -pair excitation). *For each vertex $v \in V_{600}$, define*

$$E(v) := |v - \sigma(v)|_{\text{Eucl}}^2,$$

the squared icosian-Euclidean distance between v and its σ -image, computed exactly in $K = \mathbb{Q}(\sqrt{5})$. The value lies in $\mathbb{Q}$ because the $\sqrt{5}$ -component of $|v - \sigma(v)|^2$ vanishes by σ -equivariance of the squared norm.

Lemma 3.1 (Vanishing on V_{24}). *$E(v) = 0$ if and only if $v \in V_{24}$.*

Proof. $E(v) = 0$ iff $v - \sigma(v) = 0$ iff $\sigma(v) = v$ iff $v \in V_{24}$ by Definition of the 24-cell. $\square$

Lemma 3.2 (Coordinate identity on σ -mobile vertices). *For every σ -mobile vertex $v \in V_{600} \setminus V_{24}$, the difference $v - \sigma(v)$ has exactly two nonzero coordinates, each of value $\pm\sqrt{5}/2 \in K$.*

Proof. The σ -mobile vertices are precisely the 96 even-permutation vertices, with coordinate orbit $(0, \pm 1/(2\varphi), \pm 1/2, \pm \varphi/2)$ up to even permutation. Recall the K -arithmetic identities $1/(2\varphi) = (\sqrt{5} - 1)/4$ and $\varphi/2 = (1 + \sqrt{5})/4$. Applied componentwise:

$$\begin{aligned} \sigma(0) &= 0, \\ \sigma(1/2) &= 1/2, \\ \sigma(1/(2\varphi)) &= -(1 + \sqrt{5})/4 = -\varphi/2, \\ \sigma(\varphi/2) &= (1 - \sqrt{5})/4 = -1/(2\varphi), \end{aligned}$$

so σ sends the golden coordinate values $1/(2\varphi)$ and $\varphi/2$ to their conjugate partners $(-\varphi/2$ and $-1/(2\varphi)$ respectively) in the same coordinate slots, while leaving 0 and $\pm 1/2$ slots invariant. Subtracting,

$$\begin{aligned} 1/(2\varphi) - \sigma(1/(2\varphi)) &= (\sqrt{5} - 1)/4 + (1 + \sqrt{5})/4 = \sqrt{5}/2, \\ \varphi/2 - \sigma(\varphi/2) &= (1 + \sqrt{5})/4 - (1 - \sqrt{5})/4 = \sqrt{5}/2, \end{aligned}$$

and the 0 and $\pm 1/2$ coordinates contribute zero to $v - \sigma(v)$. So exactly two coordinates of $v - \sigma(v)$ are nonzero, each of value $\pm\sqrt{5}/2$ depending on the original sign convention. $\square$

4 Main theorem: monochromatic spectrum

Theorem 4.1 (Hawking quantum). *The σ -pair excitation observable E takes exactly two values on V_{600} : $E(v) = 0$ for all 24 vertices in V_{24} and*

$$E(v) = E_q := \frac{5}{2}$$

for each of the remaining 96 σ -mobile vertices. The spectrum is the discrete distribution $24\delta_0 + 96\delta_{5/2}$.

Proof. Vanishing on V_{24} is Lemma 3.1.

For σ -mobile v , Lemma 3.2 gives $v - \sigma(v)$ having exactly two nonzero coordinates each of value $\pm\sqrt{5}/2$. Computing the squared Euclidean norm:

$$E(v) = |v - \sigma(v)|^2 = (\sqrt{5}/2)^2 + (\sqrt{5}/2)^2 = \frac{5}{4} + \frac{5}{4} = \frac{5}{2}.$$

This holds for every σ -mobile vertex regardless of which two coordinate positions are nonzero, so $E(v) = 5/2$ uniformly across all 96 such vertices.

Symmetry corroboration. Define the bilateral action Γ on V_{600} by $\Gamma := \{v \mapsto gvh : g, h \in V_{24}\}$. Each generator is a quaternionic isometry (left- and right-multiplication by unit quaternions preserve $|\cdot|^2$) and commutes with σ because $g, h \in V_{24}$ have $\sigma(g) = g$ and $\sigma(h) = h$, so $\sigma(gvh) = g\sigma(v)h$. Therefore E is constant on Γ -orbits. The Γ -orbits on V_{600} are the V_{24} -double cosets $V_{24} \cdot v \cdot V_{24}$; the certificate `verify.py` confirms that there are exactly two such orbits — V_{24} itself (of size 24) and a single orbit of size 96 (the σ -mobile sector). Constancy of E on the 96-orbit is therefore implied by Γ -invariance. $\square$

We refer to $E_q = 5/2$ as the *cascade Hawking quantum*.

5 First-law identity

Definition 5.1 (Coset thermodynamic content). *For a left coset $C = gH$ of $H = \text{Dic}_5$ in $2I$, set*

$$S(C) := |C \cap V_{24}|, \quad A(C) := |C \setminus V_{24}|, \quad E(C) := \sum_{v \in C \setminus V_{24}} E(v), \quad T_{\text{casc}}(C) := E(C)/A(C).$$

The same definitions apply to right cosets Hg .

Theorem 5.1 (Discrete first law). *For every coset C of H in $2I$ — left or right, bulk or non-bulk*

$$S(C) = 4, \quad A(C) = 16, \quad E(C) = 40, \quad T_{\text{casc}}(C) = E_q = \frac{5}{2},$$

and the following identities hold exactly:

$$E(C) = T_{\text{casc}}(C) \cdot A(C), \quad S(C) = A(C)/4, \quad T_{\text{casc}}(C) \cdot S(C) = E(C)/4.$$

Aggregating over the five non-bulk left cosets: $S_{\text{tot}} = 20$, $A_{\text{tot}} = 80$, $E_{\text{tot}} = 200$, $T_{\text{casc}} \cdot S_{\text{tot}} = E_{\text{tot}}/4 = 50$.

Proof. $S(C) = 4$ and $A(C) = 16$ are the per-coset incidence counts of Paper 1 [Smart, 2026]. By Theorem 4.1, every σ -mobile vertex contributes $E_q = 5/2$ to $E(C)$, so $E(C) = A(C) \cdot E_q = 16 \cdot 5/2 = 40$, and $T_{\text{casc}}(C) = E(C)/A(C) = 5/2$. The three identities are direct substitutions: $E(C) = T_{\text{casc}}(C) \cdot A(C)$ by definition of $T_{\text{casc}}(C)$; $S(C) = 4 = 16/4 = A(C)/4$ by Paper 1; $T_{\text{casc}}(C) \cdot S(C) = (5/2) \cdot 4 = 10 = 40/4 = E(C)/4$. The bulk coset $C = H$ itself satisfies the same per-coset identity because $|H \cap V_{24}| = 4$ (Paper 1, 24-cell incidence with H).

The aggregate values follow by summing over the five disjoint non-bulk cosets. $\square$

Remark 5.1 (Bekenstein–Hawking analogy, interpretive). *Under the interpretive identification of σ -mobile vertices as horizon channels (“area”) and σ -fixed vertices as anchor states (“entropy” bits), Theorem 5.1 produces three formal relations within this finite model that echo the structural shape of the Bekenstein–Hawking entropy formula: a sum $E = T_{\text{casc}} \cdot A$ that records E as the channel count times a common per-channel quantum, the ratio $S = A/4$ that mirrors the Bekenstein coefficient (this part is genuinely standard, by Paper 1 [Smart, 2026]), and a derived identity $T_{\text{casc}} \cdot S = E/4$ that follows from those two by direct substitution. None of these is the standard physical first law $dE = T dS$ of black-hole thermodynamics; the present finite model does not address dynamical changes. Calibration of $E_q = 5/2$ to the physical Hawking temperature $T_H^{\text{phys}} = \hbar c^3 / (8\pi G M k_B)$ is also not constructed here.*

6 Scope

Scope.

- **Established (exact, unconditional):** the σ -pair spectrum on V_{600} is $24\delta_0 + 96\delta_{5/2}$ (Theorem 4.1); the per-coset first-law identity $T_{\text{casc}} \cdot S = E/4$ with $T_{\text{casc}} = E_q = 5/2$, $A = 16$, $S = 4$, $E = 40$ (Theorem 5.1); the bilateral V_{24} symmetry orbit certificate confirms the 96-orbit structure.
- **Interpretive:** the identification of σ -mobile vertices as Hawking radiation channels and σ -fixed vertices as horizon anchor states (Remark 5.1). A different identification would attach different physical content to the same arithmetic identity.
- **Out of scope:** cascade-unit calibration to physical Planck units; semiclassical recovery of a Planck continuum from the discrete gap; quantitative primordial-black-hole evaporation cutoff; the canonical involution τ_σ (Paper 3); cosmological applications (Paper 4); ontology of V_{600} . The formal analogy between E_q and a Hawking-temperature parameter is observed without claim; the unit calibration required to lift T_{casc} from a structural finite identity to a quantitative physical Hawking temperature is not constructed here.

7 Toward a calibrated continuum limit

The finite spectrum derived here is qualitatively unlike a smooth Planck continuum. To connect the present arithmetic identity to a physical observable would require (i) a calibration that maps the cascade unit E_q to a physical energy scale, (ii) a mechanism by which sums over many cascade resolutions broaden the discrete gap into an approximately thermal distribution, and (iii) an observational model that distinguishes a discrete cutoff from a thermal tail in any candidate setting (for instance, late-stage primordial black-hole evaporation [Carr et al., 2020]). None of (i)–(iii) is constructed here, and we make no quantitative observational claim. We note the qualitative shape of the gap as a target for future calibration work.

8 Conclusion

The σ -Galois pair-excitation observable on the binary icosahedral group $2I$ admits a closed-form spectrum: 24 zero modes (the σ -fixed 24-cell sublattice) and 96 mobile modes at the single quantum value $E_q = 5/2$ (Theorem 4.1). Combined with the Paper 1 coset-incidence identity, this produces a per-coset first-law identity $T_{\text{casc}} \cdot S = E/4$ with $T_{\text{casc}} = E_q$ (Theorem 5.1). The result is exact, parameter-free, and verifiable by the accompanying script. We frame this as a discrete arithmetic analogue of the Hawking-radiation thermodynamic structure, deferring physical-unit calibration and semiclassical-continuum recovery to future work.

A Verification certificate

The companion script `verify.py` performs the following exact checks and exits with status 0 when all assertions hold:

1. $|V_{600}| = 120$, $|V_{24}| = 24$, $|V_{600} \setminus V_{24}| = 96$.
2. For all 24 σ -fixed vertices, $E(v) = 0$ exactly.
3. For all 96 σ -mobile vertices, $E(v) = 5/2$ exactly, with the rational value computed in $\mathbb{Q}(\sqrt{5})$.
4. Coordinate proof: for each of the 96 σ -mobile vertices, the difference $v - \sigma(v)$ has exactly two nonzero coordinates with rational part 0 and $\sqrt{5}$ -part $\pm 1/2$.
5. Bilateral V_{24} symmetry: σ commutes with left-multiplication by every $g \in V_{24}$ (24 generators), and the bilateral action $\Gamma = \{v \mapsto gvh : g, h \in V_{24}\}$ on V_{600} has exactly two orbits — V_{24} (size 24) and the σ -mobile sector (size 96).
6. Per-coset first-law identity for all 6 left cosets and all 6 right cosets (bulk and non-bulk): $A = 16$, $S = 4$, $E = 40$, $T_{\text{casc}} = 5/2$, $T_{\text{casc}} \cdot S = E/4 = 10$.
7. Aggregate first law: $A_{\text{tot}} = 80$, $S_{\text{tot}} = 20$, $E_{\text{tot}} = 200$, $T_{\text{casc}} \cdot S_{\text{tot}} = E_{\text{tot}}/4 = 50$.

The script depends only on the Python standard library and the shared `vfd_v600` package (`papers/v600-programme/lib`). Reproducibility command:

```
PYTHONPATH=papers/v600-programme/lib \
python3 papers/v600-hawking-quantum/verify.py
```

References

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